Adaptive immunity

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SGL4

Objectives:

- To know the following topics:
- Adaptive immunity classified into passive and active.
- There are 2 Arms of adaptive immune responses humoral and cellular.
- The students must learn all detail about the definition, types
,and arms of immunity.

Types of Immunity

- There are 2 types of immunity:
- 1-innate immunity
- 2-Adaptive immunity.
- Innate immunity includes mechanical, chemical, and biological factors.
- Adaptive immunity has many features, and it has Humerol and cellular arms.

Adaptive Immunity:-

It is a type of immune response that is stimulated by exposure to infectious agents.

- It improves upon repeated exposure.
- Have memory.
- MHC restricted.

2-Adaptive immunity:

- It has capacity to distinguish between different, even closely related microbs and molecules, so it is called **specific immunity.**

- Because this form of immunity develops as a response to infection, it is called **adaptive immunity**.

Classification of adaptive immunity

- **Adaptive immunity classified**

into 1-Active adaptive immunity:-

- **A-**Natural active acquired immunity (infection)
- **B-**Artificial active acquired immunity (vaccination)

2-Passive adaptive immunity:-

- **A-**Natural passive adaptive immunity (colostrums)

Arms of adaptive immune responses:

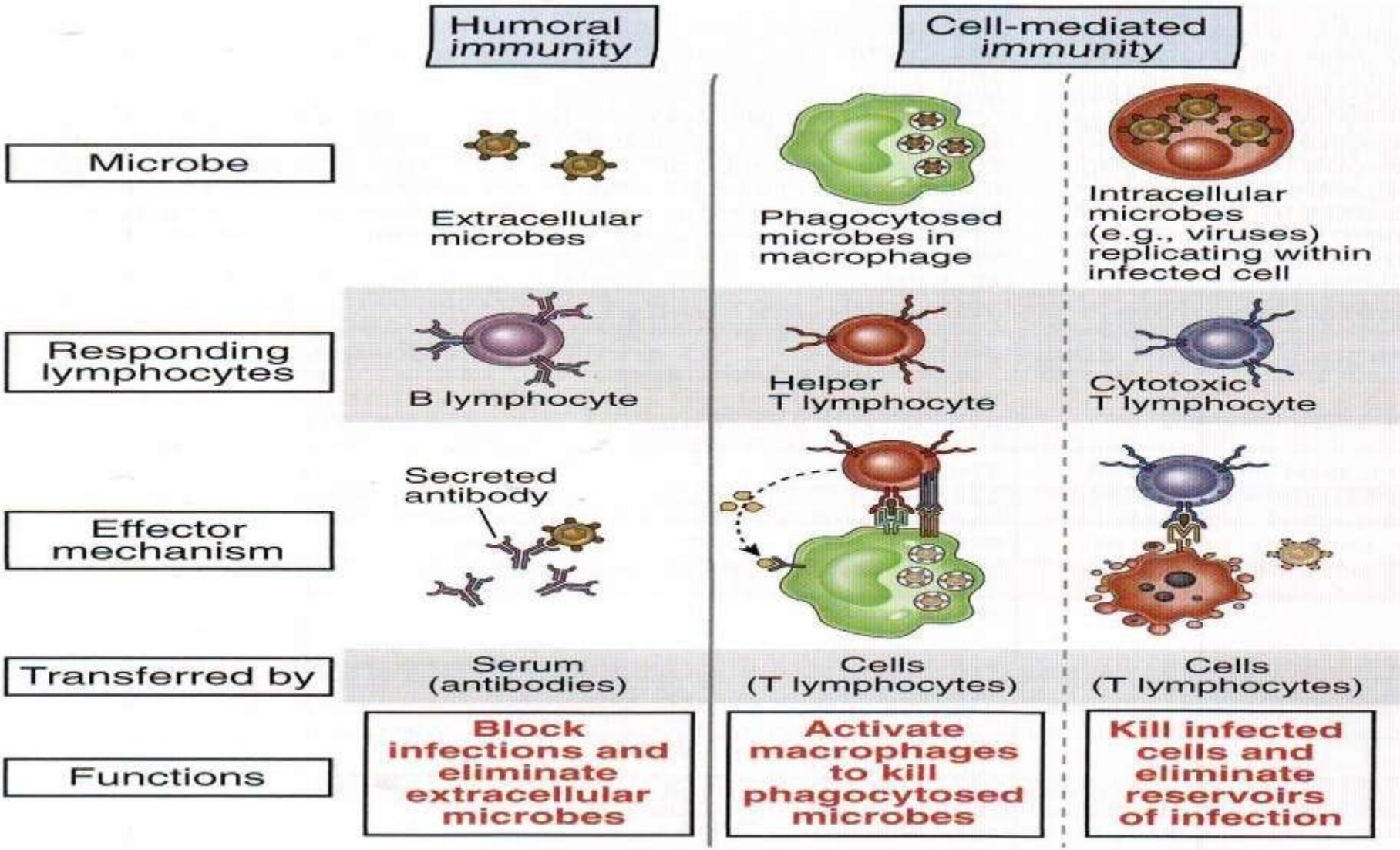
1 **Humoral immunity:**

- It is mediated by molecules in the blood and mucosal secretions called antibodies(abs), which are produced by B lymphocytes(B cells), this abs recognize microbial antigens and neutralize the infectivity of the microbes.
- It is effective against extracellular microbes and their toxins.
- This type of immune response mediated by T helper 2(Th2) lymphocyte.

Arms of adaptive immune responses:

2-Cell mediated immunity(CMI):

- It is mediated by T helper 1(Th1) lymphocyte and, T cytotoxic(TC)
- It is effective against intracellular microbes like a virus, many fungal infections and some bacteria like TB, survive and proliferate inside phagocyte.
- Also it has a role in transplantation and tumor immunity.



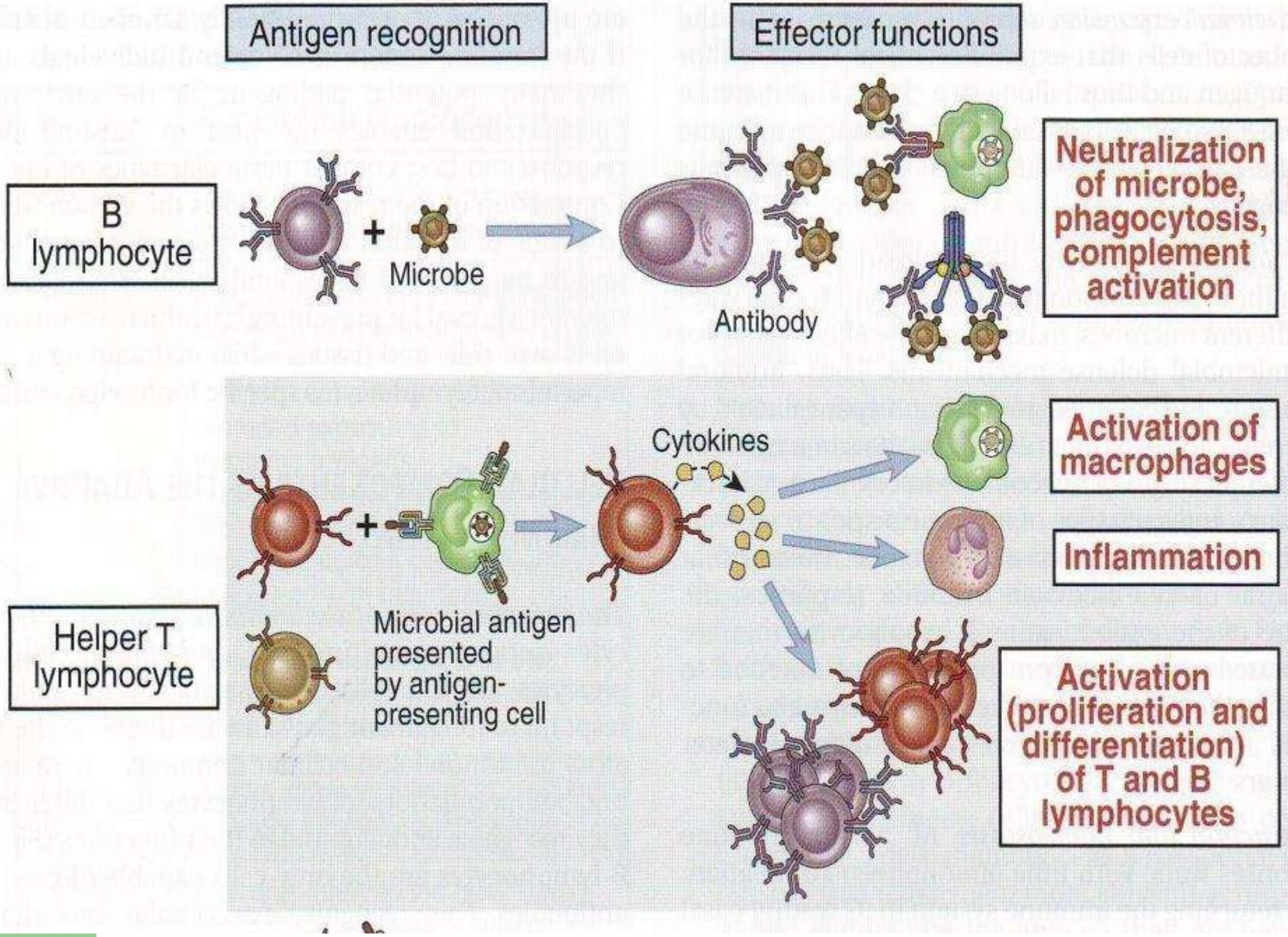
B lymphocytes secrete abs to eliminate extracellular pathogen.
T helper cells activate macrophage to kill intracellular pathogen.
T cytotoxic directly kills infected cells. (Abbas, 2007)

Cellular component of the adaptive immune response:

- **Lumphocytes:** The cells that specifically recognize and respond to the foreign antigen so they mediate humoral and CMI.
- **B cells:** Recognize extracellular antigens and differentiate to plasma cells that produce abs as humoral immunity.

Cellular component of the adaptive immune response:

- **Phagocytes:** Ingest microbes and kill them, Helper T cells enhance the microbicidal abilities of the phagocytes.
- **T cells:** Recognize the only antigen attached to MHC which expressed on the surfaces of other cells. Also, they recognize intracellular antigens.
- **Cytotoxic T cells:** Destroy cells infected by microbes that are inaccessible to antibodies.


Antigen recognition
Effector functions
B lymphocyte
Microbe
Antibody
Neutralization of microbe, phagocytosis, complement activation
Cytokines
Activation of macrophages
Inflammation
Helper T lymphocyte
Microbial antigen presented by antigen-presenting cell
Activation (proliferation and differentiation) of T and B lymphocytes

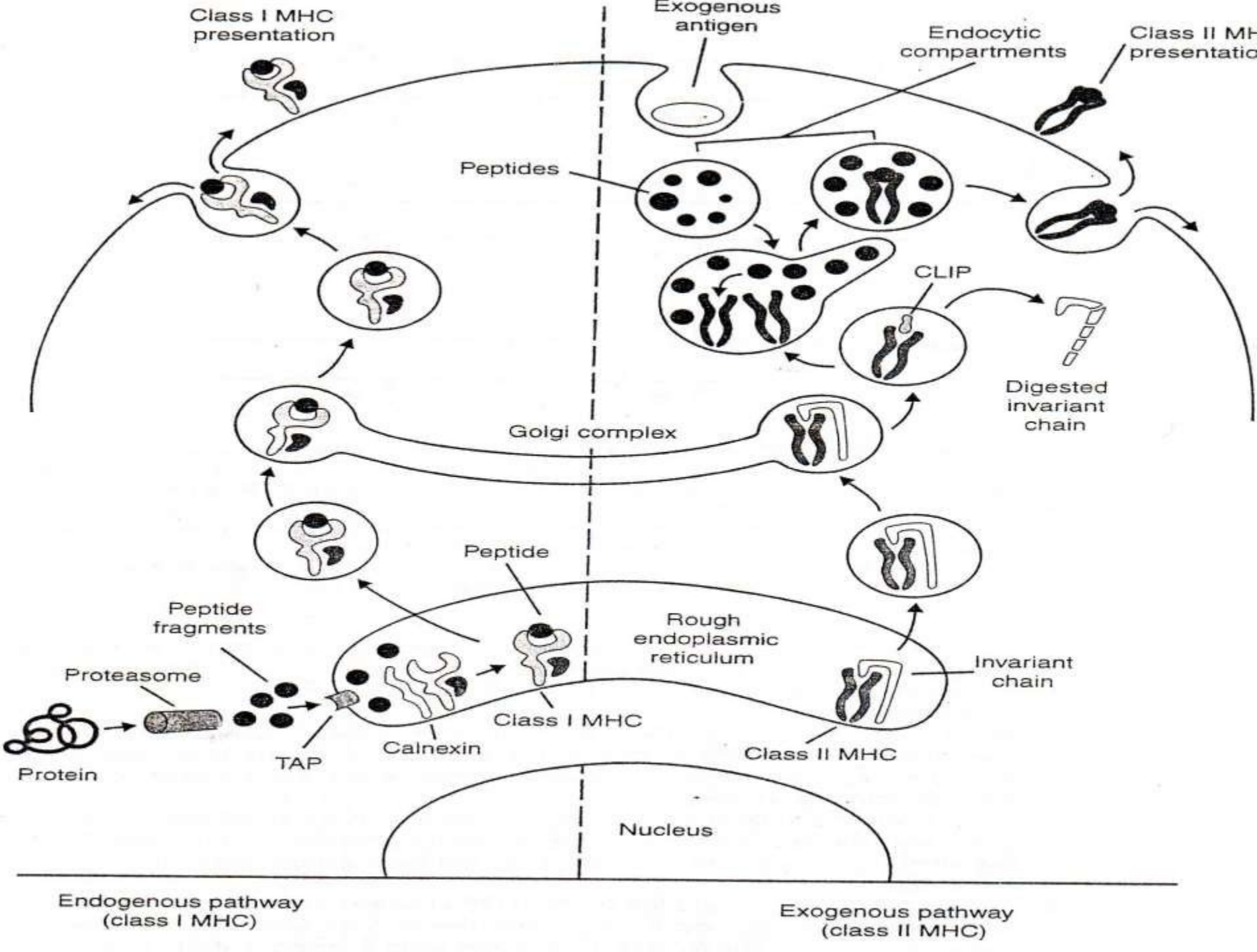
(Abbas, 2007)

General steps of the immune response:-

- **A- Processing:-**
- Processing of antigen within the cytoplasm of macrophage into small peptides that associate

the class II MHC proteins or class I MHC protein

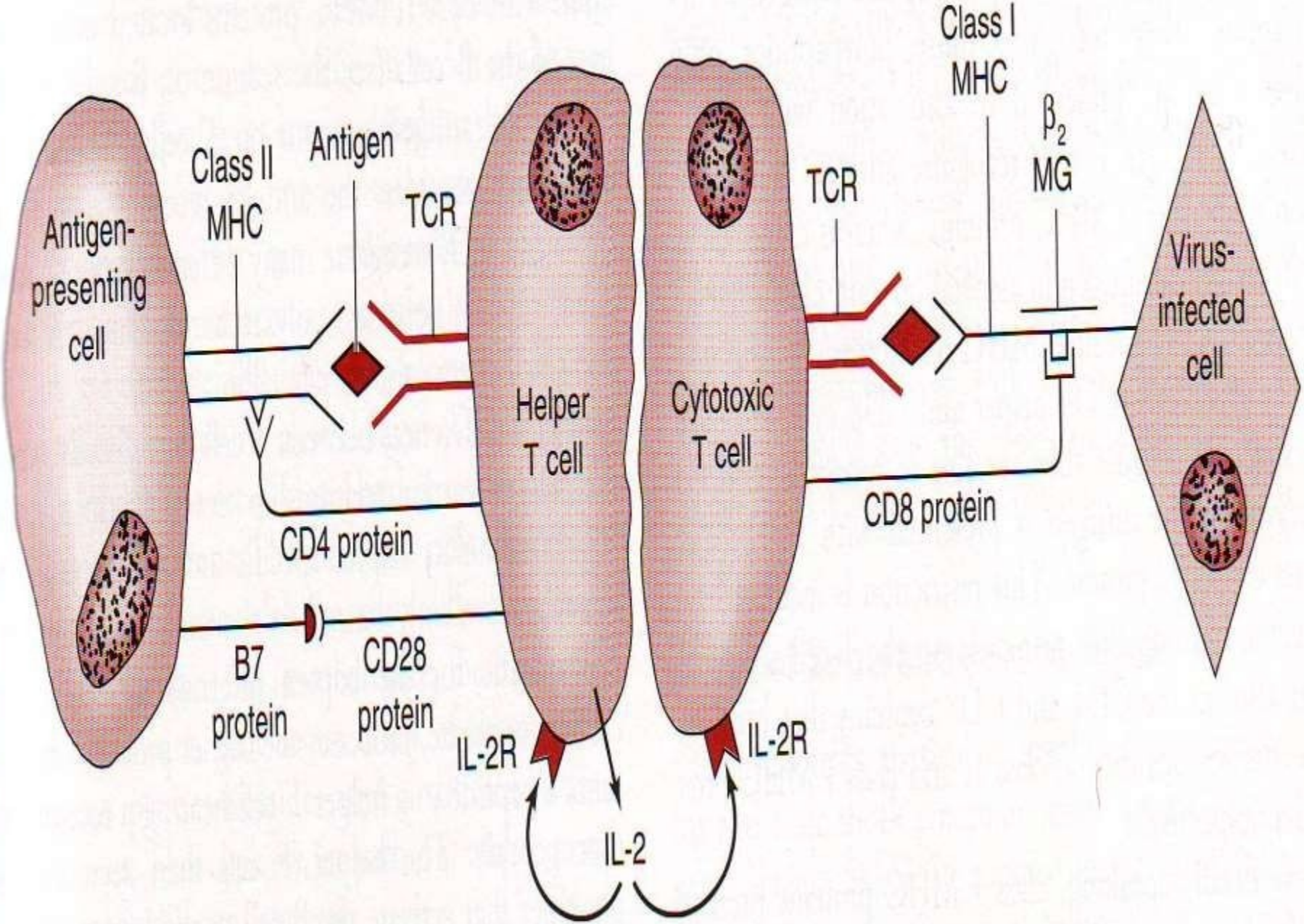
- **B- Transported and presentation:- The complex is transported to the surface of the macrophage then presented to the receptor on the CD4 positive helper cell.** On the other hand, viral infected cell presented the antigen associated with MHC class I and presented it to receptor on CD8 positive cytotoxic cell.

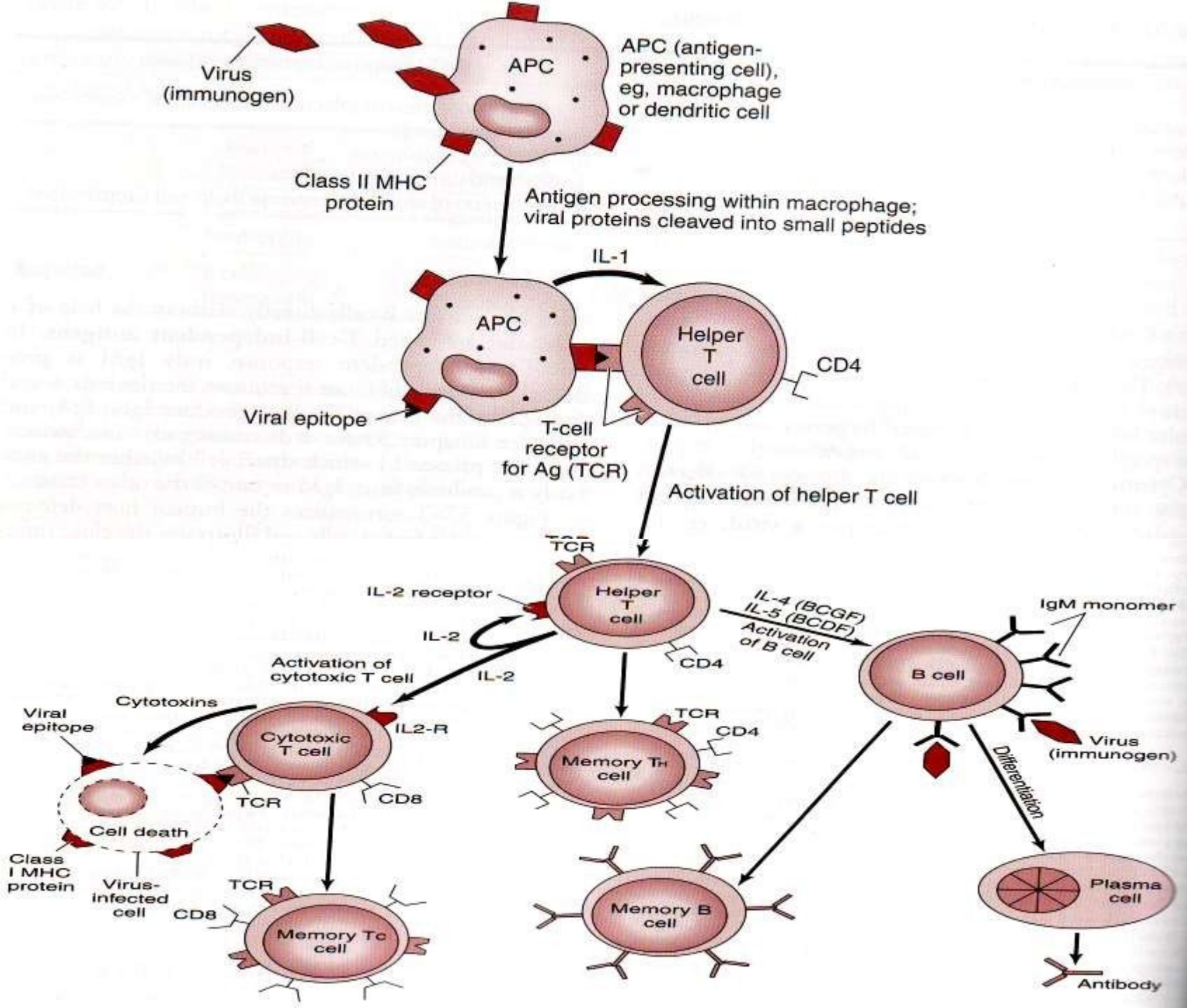

Class I MHC presentation
Exogenous antigen
Endocytic compartments
Class II MH presentatio
Peptides
CLIP
Digested invariant chain
Golgi complex
Peptide
Peptide fragments
Rough endoplasmic reticulum
Invariant chain
Proteasome
Class I MHC
Calnexin
TAP
Class II MHC
Protein
Nucleus
Endogenous pathway (class I MHC)
Exogenous pathway (class II MHC)

Signals activate T cells:

1- Interaction of the complex (processed antigen and MHC protein) with the T-cell receptor. This combination occurs by LFA-1 (lymphocyte function-associated antigen) protein on T-cells binds to ICAM-1(intracellular adhesion molecule) protein on APCs. Indeed, B7 protein on the APC interacts with CD28 protein on the helper T cell.

- 2- Interleukin-1 produced by the macrophage is necessary for efficient helper T cell activation.
- Helper T cell produces IL-2 as well as the IL-2 receptor.
- IL-2 also known as T- cell growth factor stimulates the helper T-cell to multiply into a clone of antigen-specific helper T cell. Most cells of this clone perform effectors and regulatory functions, but some become memory cells.


Class I
MHC
β_2
MG
Class II
MHC
Antigen
TCR
TCR
Antigen-
presenting
cell
Virus-
infected
cell
Helper
T cell
Cytotoxic
T cell
CD4 protein
CD8 protein
B7
protein
CD28
protein
IL-2R
IL-2R
IL-2


Virus (immunogen)
APC
APC (antigen-presenting cell), eg, macrophage or dendritic cell
Class II MHC protein
Antigen processing within macrophage; viral proteins cleaved into small peptides
IL-1
APC
Helper T cell
CD4
Viral epitope
T-cell receptor for Ag (TCR)
Activation of helper T cell
TCR
IL-2 receptor
Helper T cell
IL-4 (BCGF)
IL-5 (BCDF)
Activation of B cell
IgM monomer
IL-2
CD4
Activation of cytotoxic T cell
IL-2
B cell
Cytotoxins
Viral epitope
Cytotoxic T cell
IL2-R
TCR
CD4
Memory TH cell
Virus (immunogen)
TCR
CD8
Cell death
Class I MHC protein
Virus-infected cell
TCR
CD8
Memory Tc cell
Memory B cell
Differentiation
Plasma cell
Antibody

Summary

- There are 2 types of immunity innate and adaptive.
- Innate immunity includes mechanical, chemical, and biological factors.
- **Adaptive immunity** is a type of immune responses that are stimulated by exposure to infectious agents.
- It improves upon repeated exposure, Has memory and MHC restricted.Adaptive immunity **classified** into active and passive.
- Arms of adaptive immune responses are humoral and cellular.

References

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Thanks a lot